

LIBID_LibName

Function block library for PLCnext Engineer

Library version: 2.0.0

Document revision: 2.20

Release date: 2026-04-30

This documentation is available in English only.

Your Company

Company Street 123
12345 City, Country

CONTENTS

1. General information
2. Installation hints
3. Change notes
4. Naming conventions
5. Library components
6. LIBID_My_FU1_1
7. LIBID_My_FB1_1
8. Group 1
 - LIBID_My_FB2_1
 - LIBID_My_FB3_1
9. HMI symbols
 - LIBID_SYM_Name_1
10. User-defined data types
11. Diag codes of used firmware blocks
12. Third-party licenses
13. Known issues
14. Startup example(s)
15. Support

1 General information

1.1 Library description

Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. At vero eos et accusam et justo duo dolores et ea rebum. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet. Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. At vero eos et accusam et justo duo dolores et ea rebum. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet.

1.2 Supported PLCnext Engineer versions

Minimum required version:

Maximum supported version:

1.3 Supported PLCs and firmware

Product category	Item designation	Item no.	Firmware
Controller	AXC F 1152	1151412	2025.0 LTS, 2025.0.x
Controller	AXC F 2152	2404267	2026.0 LTS, 2026.0.x
Safety controller	RFC 4072S	1051328	2026.3

1.4 Safety-related danger messages

In accordance with the applicable standards, the following hazard messages are used in this documentation.

Note

Note texts give you additional information that may be useful when working with this library.

Caution

A caution text informs you about a minor hazard situation which, if not avoided, may result in minor or moderate injury.

Warning

This is a warning text. It informs you about a hazardous situation which, if not avoided, could result in death or serious injury.

Danger

This text indicates a hazardous situation which, if not avoided, will result in death or serious injury.

2 Installation hints

If you did not specify a different directory during library installation, all data in the MSI file will be unpacked to

C:\Users\Public\Documents\Phoenix Contact Libraries\PLCnext Engineer
(former: PC Worx Engineer)

Please copy the library data to the working library directory of PLCnext Engineer.

If you did not specify a different directory during PLCnext Engineer installation, the default PLCnext Engineer working library directory is

C:\Users\Public\Documents\PLCnext Engineer\Libraries
(former: PC Worx Engineer\Libraries)

Note

If HMI objects from this library are used in your project, make sure that English is configured as the default language:

PLCnext Engineer > Extras > Options > Default Language Settings > en

3 Change notes

Version	Build date	Tested with PLCnext Engineer	New	Fixed
2.0.0	2026-03-04	2026.0 LTS	LIBID_My_FU1_1, LIBID_My_FB4_1	Data type mismatch in LIBID_My_FB1_1
1.1.0	2025-09-10	2025.6	LIBID_My_FB1_1 with further diagnostic information	
1.0.0	2025-02-26	2025.0 LTS	Initial release	

* **Library version format:** Major.Minor.Patch

4 Naming conventions

4.1 Prefixes for the usage types of variables (scope)

Designation	PLCopen	In this library
Global flag	g	<LIBID>_G
Process data input	i	<LIBID>_I
Process data output	o	<LIBID>_Q
Local variable	l	none
Function / function block input parameter	p	IN
Function block output parameter	p	OUT
Function block in/out parameter	p	INOUT
Program-global variable	n. d.	<LIBID>_Gp
Instance-global input	n. d.	<LIBID>_Ii
Instance-global output	n. d.	<LIBID>_Qi
Global HMI variable	n. d.	<LIBID>_HMI
Local HMI variable	n. d.	hmi
Input of a method	n. d.	mIN
Local variable of a method	n. d.	m
IN Port (GDS)	n. d.	INPORT
OUT Port (GDS)	n. d.	OUTPORT
Safe process data input	see table below	<LIBID>_SI
Safe process data output	see table below	<LIBID>_SQ
Variable for data exchange between safety and non-safety programming	see table below	<LIBID>_SX
Local safe variable	see table below	s
Safe function block input	see table below	SIN
Safe function block output	see table below	SOUT

* n. d. - not defined

4.2 Prefixes for data types of variables

Designation	PLCopen	In this library
BOOL	x	x
SINT	si	si
INT	i	i
DINT	di	di
LINT	li	li
USINT	usi	usi
UINT	ui	ui
UDINT	udi	udi
ULINT	uli	uli
REAL	r	r
LREAL	lr	lr
TIME	tim	t
LTIME	ltim	lt
DATE	dt	d
LDATE	ldt	ldt
TIME_OF_DAY / TOD	tod	td
LTIME_OF_DAY / TOD	ltod	ltd
DATE_AND_TIME / DT	dt	dt
LDATE_AND_TIME / DT	ldt	ldt
STRING	str	str
WSTRING	wstr	wstr
BYTE	by	b
WORD	w	w
DWORD	dw	dw
LWORD	lw	lw
Safe data types	Data type prefix ends with: s; Example: SAFEBOOL prefix is: xs	Differentiation is at scope-level, not for data type

4.3 Suffixes for variables

Designation	PLCopen	In this library
Normally closed	n. d.	NC

* n. d. - not defined

4.4 Exemplary variable names

- Local boolean variable: xVarName
- Global flag of data type REAL: <LIBID>_G_rVarName
- Variable for a normally closed switch: <LIBID>_I_xSwitch_NC
- Local HMI variable of data type INT: hmi_iVarName
- Function block input of data type TIME: IN_tVarName
- Safe local boolean variable: s_xVarName

4.5 Prefixes for data type definitions

Designation	PLCopen	In this library
Structure	ST	<LIBID>_ST
Enumeration	E	<LIBID>_E
Array	A	<LIBID>_ARR
User-defined STRING (WSTRING)	n. d.	<LIBID>_(W)STR_<length>

* n. d. - not defined

4.6 Exemplary names for data types

- Structure: <LIBID>_ST_Name
- Enumeration: <LIBID>_E_Name
- Array with fields from 1 to 100 of data type DINT: <LIBID>_ARR_DI_1_100
- User-defined STRING with 45 characters: <LIBID>_STR_45

4.7 Prefixes for component types

Designation	PLCopen	In this library
Program	PRG	<LIBID>_PG
Function block	FB	<LIBID>_FB
Function	FU	<LIBID>_FU
Method	n. d.	<LIBID>_MTH
SFC step name	n. d.	<LIBID>_S
SFC action (worksheet)	n. d.	<LIBID>_A
SFC transition (worksheet)	n. d.	<LIBID>_T
SFC interlock (worksheet)	n. d.	<LIBID>_LCK
HMI page	n. d.	<LIBID>_PAGE
HMI symbol	n. d.	<LIBID>_SYM

* n. d. - not defined

4.8 Exemplary names for components

- Function block version 2: <LIBID>_FB_Name_2
- Method version 1: <LIBID>_MTH_Name_1
- HMI symbol version 3: <LIBID>_SYM_Name_3

5 Library components

5.1 Functions

Name	Version	Short description	Related products
LIBID_My_FU1_1	1.0	This function can be used to ... Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. At vero eos et accusam et justo duo dolores et ea rebum. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet.	/

5.2 Function blocks

Name	Version	Short description	Related products
LIBID_My_FB1_1	1.0	This function block can be used to ...	AXL SE RS232 (item no. 1181787)
LIBID_My_FB2_1	1.0	This function block can be used to ...	/
LIBID_My_FB3_1	1.0	This function block can be used to ...	AXL SE RS232 (item no. 1181787)

5.3 HMI symbols

Name	Version	Short description	Related products
LIBID_SYM_Name_1	1.0	This HMI symbol can be used to ...	/

6 LIBID_My_FU1_1

Type: Function

6.1 Description

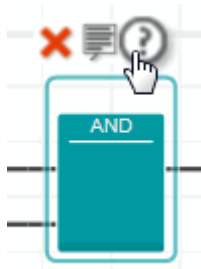
This function is used to

The function provides the following:

- Item 1
- Item 2
- ...

6.2 Example(s)

Example in FBD:



Note

- All parameters can be negated.

Negation of formal parameters is not supported in safety-related code (SNOLD).

- ...

6.3 Inputs

Variable	Data type	Description
IN_xActivate	BOOL	Function activation
IN_xVar1	BOOL	Description of parameter xVar1
IN_xVar2	BOOL	Description of parameter xVar2

6.4 Return value

Data type	Description
BOOL	Description of the return value

6.5 Additional information

This function can be combined with the HMI symbol LIBID_SYM_Name_1.

7 LIBID_My_FB1_1

Type: Function block

7.1 Description

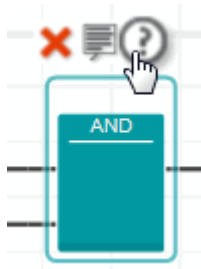
This function block is used to ...

The function block provides the following:

- Item 1
- Item 2
- ...

7.2 Example(s)

Example in FBD:



Note

- All parameters can be negated.

Negation of formal parameters is not supported in safety-related code (SNOLD).

- Function blocks must be instantiated. The instance name of the function block has to be declared in the 'Variables' table of the POU where the FB is going to be used. The instance name must be unique within the POU.
- ...

7.3 Inputs

Variable	Data type	Description
IN_xActivate	BOOL	Function block activation
IN_bVar1	BYTE	Description of parameter bVar1
IN_eColor	LIBID_E_TrafficLight	Enumeration used for color selection

7.4 Outputs

Variable	Data type	Description
OUT_xActive	BOOL	Function block active status
OUT_xError	BOOL	Function block error status
OUT_wDiagCode	WORD	Diagnostic code. See the “Error codes” section for further information
OUT_wAddDiagCode	WORD	Additional diagnostic code. See the “Error codes” section for further information
OUT_rResult	REAL	Description of parameter rResult

7.5 InOut

Variable	Data type	Description
INOUT_stMotorData	LIBID_ST_Motor	Motor data
INOUT_arrAllMotorData	LIBID_ARR_Motor_0_6	Collected motor data of all motors

7.6 Dependencies

Global variables:

Name	Data type	Description
RTC	RTC_TYPE	Real-time clock (date and time of the PLC)

Library-internal and external functions and function blocks:

Name	Source	Description
PTD_FB_Zones_1	PTD_ParcelTurning_1_3_1	Outputs drive zone/position
LIBID_My_FU1_1	Library-internal	Short description of LIBID_My_FU1_1

7.7 Error codes

Error codes for OUT_wDiagCode:

OUT_wDiagCode	Description
16#0000	Function block is deactivated
16#8000	Function block is in regular operation
...	...

Error codes for OUT_wDiagCode and OUT_wAddDiagCode:

OUT_wDiagCode	OUT_wAddDiagCode	Description
16#0000	16#0000	Function block is deactivated
16#8000	16#0000	Function block is in regular operation
16#C110		An invalid input parameter has been detected
16#C520		Error in an internally used function block
...	...	...

7.8 Additional information

This function block can be combined with the HMI symbol LIBID_SYM_Name_1.

Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. At vero eos et accusam et justo duo dolores et ea rebum. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet. Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. At vero eos et accusam et justo duo dolores et ea rebum. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet. Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. At vero eos et accusam et justo duo dolores et ea rebum. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet.

8 Group 1

In this group you can find function blocks for ...

List of function blocks:

- LIBID_My_FB2_1
- LIBID_My_FB3_1
- ...

8.1 LIBID_My_FB2_1

Type: Function block

8.1.1 Description

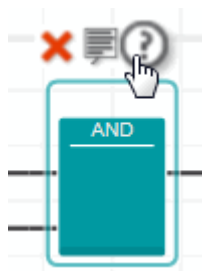
This function block is used to ...

The function block provides the following:

- Item 1
- Item 2
- ...

8.1.2 Example(s)

Example in FBD:



Note

- All parameters can be negated.

Negation of formal parameters is not supported in safety-related code (SNOLD).

- Function blocks must be instantiated. The instance name of the function block has to be declared in the 'Variables' table of the POU where the FB is going to be used. The instance name must be unique within the POU.
- ...

8.1.3 Inputs

Variable	Data type	Description
IN_xActivate	BOOL	Function block activation
IN_bVar1	BYTE	Description of parameter bVar1

8.1.4 Outputs

Variable	Data type	Description
OUT_xActive	BOOL	Function block active status
OUT_xError	BOOL	Function block error status
OUT_wDiagCode	WORD	Diagnostic code. See the “Error codes” section for further information
OUT_wAddDiagCode	WORD	Additional diagnostic code. See the “Error codes” section for further information
OUT_rResult	REAL	Description of parameter rResult

8.1.5 InOut

Variable	Data type	Description
INOUT_stMotorData	LIBID_ST_Motor	Motor data

8.1.6 Dependencies

Global variables:

Name	Data type	Description
RTC	RTC_TYPE	Real-time clock (date and time of the PLC)

Library-internal and external functions and function blocks:

Name	Source	Description
PTD_FB_Zones_1	PTD_ParcelTurning_1_3_1	Outputs drive zone/position
LIBID_My_FU1_1	Library-internal	Short description of LIBID_My_FU1_1

8.1.7 Error codes

Error codes for OUT_wDiagCode:

OUT_wDiagCode	Description
16#0000	Function block is deactivated
16#8000	Function block is in regular operation
...	...

Error codes for OUT_wDiagCode and OUT_wAddDiagCode:

OUT_wDiagCode	OUT_wAddDiagCode	Description
16#0000	16#0000	Function block is deactivated
16#8000	16#0000	Function block is in regular operation
16#C110		An invalid input parameter has been detected
16#C520		Error in an internally used function block
...	...	...

8.1.8 Additional information

This function block can be combined with the HMI symbol LIBID_SYM_Name_1.

8.2 LIBID_My_FB3_1

Type: Function block

8.2.1 Description

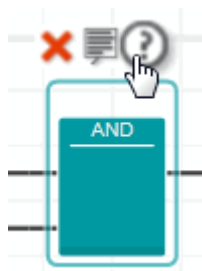
This function block is used to ...

The function block provides the following:

- Item 1
- Item 2
- ...

8.2.2 Example(s)

Example in FBD:



Note

- All parameters can be negated.

Negation of formal parameters is not supported in safety-related code (SNOLD).

- Function blocks must be instantiated. The instance name of the function block has to be declared in the 'Variables' table of the POU where the FB is going to be used. The instance name must be unique within the POU.
- ...

8.2.3 Inputs

Variable	Data type	Description
IN_xActivate	BOOL	Function block activation
IN_bVar1	BYTE	Description of parameter bVar1

8.2.4 Outputs

Variable	Data type	Description
OUT_xActive	BOOL	Function block active status
OUT_xError	BOOL	Function block error status
OUT_wDiagCode	WORD	Diagnostic code. See the “Error codes” section for further information
OUT_wAddDiagCode	WORD	Additional diagnostic code. See the “Error codes” section for further information
OUT_rResult	REAL	Description of parameter rResult

8.2.5 InOut

Variable	Data type	Description
INOUT_stData	LIBID_ST_Motor	Motor data

8.2.6 Dependencies

Global variables:

Name	Data type	Description
RTC	RTC_TYPE	Real-time clock (date and time of the PLC)

Library-internal and external functions and function blocks:

Name	Source	Description
PTD_FB_Zones_1	PTD_ParcelTurning_1_3_1	Outputs drive zone/position
LIBID_My_FU1_1	Library-internal	Short description of LIBID_My_FU1_1

8.2.7 Error codes

Error codes for OUT_wDiagCode:

OUT_wDiagCode	Description
16#0000	Function block is deactivated
16#8000	Function block is in regular operation
...	...

Error codes for OUT_wDiagCode and OUT_wAddDiagCode:

OUT_wDiagCode	OUT_wAddDiagCode	Description
16#0000	16#0000	Function block is deactivated
16#8000	16#0000	Function block is in regular operation
16#C110		An invalid input parameter has been detected
16#C520		Error in an internally used function block
...	...	...

8.2.8 Additional information

My additional information ...

9 HMI symbols

In this group you can find HMI symbols.

List of HMI symbols:

- LIBID_SYM_Name_1
- ...

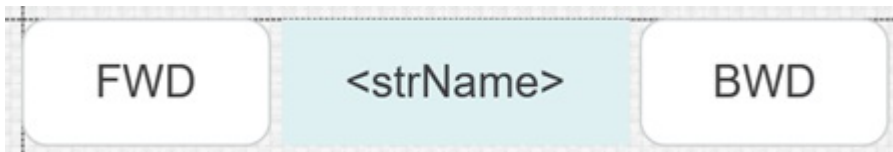
9.1 LIBID_SYM_Name_1

Type: HMI symbol

9.1.1 Description

This HMI symbol is used to

9.1.2 Example(s)



Note

- HMI symbols must be instantiated. An instance of an HMI symbol can be created either within another symbol or on an HMI page.
- A symbol may be instantiated multiple times; however, each instance must have a unique object ID.
- ...

9.1.3 HMI parameters

Parameter name	Type	UDT	Description	Input/output for symbol
strName	ANY_STRING		Device name to be displayed	IN
xMoveBwdReq	BOOL		Request to drive backwards via button	IN
xMoveFwdReq	BOOL		Request to drive forward via button	OUT
xRearPos_NC	BOOL		Sensor at rear position (normally closed)	OUT
xFrontPos_NC	BOOL		Sensor at front position (normally closed)	OUT

9.1.4 Dependencies

This HMI symbol requires the instantiation of LIBID_My_FB1_1.

9.1.5 Additional information

My additional information ...

10 User-defined data types

10.1 Structure LIBID_ST_Motor

Element name	Data type	Initial value	Description
xMotorStart	BOOL	FALSE	Trigger to start motor
xMotorStop	BOOL	FALSE	Trigger to stop motor
iSpeed	INT	0	Motor speed
xError	BOOL	FALSE	Motor error occurred

10.2 Enumeration LIBID_E_TrafficLight

Element name	Value	Is initial value	Description
Red	0	x	Red traffic light
Yellow	1		Yellow traffic light
Green	5		Green traffic light

10.3 Arrays of elementary data types

Array name	Base-type	Lower bound	Upper bound
LIBID_ARR_W_1_8	WORD	1	8
LIBID_ARR_DI_0_20	DINT	0	20
LIBID_ARR_UI_0_20	UINT	0	20
LIBID_ARR_R_0_25	REAL	0	25

10.4 Arrays of project data types

Array name	Base-type	Lower bound	Upper bound
LIBID_ARR_Motor_0_6	LIBID_ST_Motor	0	6

11 Diag codes of used firmware function blocks

11.1 PDI_READ

ERROR = TRUE

STATUS[0]	STATUS[1]	Meaning
16#09B0	16#000C	The variable connected to RD_1 is invalid (no array or invalid array type).
16#09B0	16#000B	The array connected to RD_1 is too small to save the requested receive data.
16#09B0	16#000E	Timeout. No response to the sent PDI READ request received.
16#09B0	16#000F	An internal error has occurred.

When receiving a negative confirmation as response to a PDI_READ request, the Axioline module directly copies the received error code (Error_Code and Add_Info) to STATUS[0] or STATUS[1]. These error codes are module-specific. For a description see the respective module documentation.

11.2 PDI_WRITE

ERROR = TRUE

STATUS[0]	STATUS[1]	Meaning
16#09B0	16#000A	The variable connected to SD_1 is invalid (no array or invalid array type). Connected to SD_1, greater than the maximum allowed length (245 bytes) or equal to zero.
16#09B0	16#0009	Invalid value at DATA_CNT input. The value is greater than the array size.
16#09B0	16#000E	Timeout. No response to the sent PDI WRITE request received.
16#09B0	16#000F	An internal error has occurred.

When receiving a negative confirmation as response to a PDI_WRITE request, the Axioline module directly copies the received error code (Error_Code and Add_Info) to STATUS[0] or STATUS[1]. These error codes are module-specific. For a description see the respective module documentation.

11.3 RDREC

Error code	Meaning
16#0000	No error occurred.
16#F001	Too many instances used.
16#F002	Error during initialization of the function block.
16#F003	Invalid ID.
16#F004	Invalid HANDLE/ID.
16#F005	Resources conflict.
16#F006	A function block internal task could not be generated.
16#F007	Too many instances used.
16#F008	Invalid type of a parameter.
16#F009	Invalid parameter value.
16#F00A	Invalid parameter.
16#F00B	Invalid length specified.
16#F00C	ID could not be created (too many IDs).
16#F00D	No entry found that matches the specified ID.
16#F00F	No further entries found.
16#F010	Entry in use.
16#F011	Alarm acknowledgment could not be done.
16#F012	Error reading the AR parameters (1st time).
16#F013	Negative acknowledgment received for the execution of a PROFINET service.
16#F014	Invalid length for parameter LEN/MLEN or/and RECORD data record too short.
16#F015	The service used to read the RECORD data record could not be run.
16#F016	The service used to write the RECORD data record could not be run.
16#F017	Service acknowledgment not received.
16#F018	Invalid INDEX used to access the RECORD data record of the IO device, for example, INDEX greater than 16#7FFF.
16#F019	Unknown command code.
16#F01A	Error starting the Application Relation (AR).
16#F01B	Error stopping the Application Relation (AR).
16#F01C	Notification of stopped Application Relation (AR) failed.
16#F01D	Setting the "Drive BF" flag failed.
16#F01E	Error reading the AR parameters (2nd time).

11.4 WRREC

Error code	Meaning
16#0000	No error occurred.
16#F001	Too many instances used.
16#F002	Error during initialization of the function block.
16#F003	Invalid ID.
16#F004	Invalid HANDLE/ID.
16#F005	Resources conflict.
16#F006	A function block internal task could not be generated.
16#F007	Too many instances used.
16#F008	Invalid type of a parameter.
16#F009	Invalid parameter value.
16#F00A	Invalid parameter.
16#F00B	Invalid length specified.
16#F00C	ID could not be created (too many IDs).
16#F00D	No entry found that matches the specified ID.
16#F00F	No further entries found.
16#F010	Entry in use.
16#F011	Alarm acknowledgment could not be done.
16#F012	Error reading the AR parameters (1st time).
16#F013	Negative acknowledgment received for the execution of a PROFINET service.
16#F014	Invalid length for parameter LEN/MLEN or/and RECORD data record too short.
16#F015	The service used to read the RECORD data record could not be run.
16#F016	The service used to write the RECORD data record could not be run.
16#F017	Service acknowledgment not received.
16#F018	Invalid INDEX used to access the RECORD data record of the IO device, for example, INDEX greater than 16#7FFF.
16#F019	Unknown command code.
16#F01A	Error starting the Application Relation (AR).
16#F01B	Error stopping the Application Relation (AR).
16#F01C	Notification of stopped Application Relation (AR) failed.
16#F01D	Setting the "Drive BF" flag failed.
16#F01E	Error reading the AR parameters (2nd time).

11.5 TCP_SOCKET, TCP_READ, TCP_WRITE, UDP_SOCKET, UDP_READ, UDP_WRITE

ERROR = FALSE

Error code	Meaning
16#0000	Situation is normal (no error).
16#8000	Socket is attempting to connect to the partner.
16#8001	Server is listening for a client.
16#8002	Server has rejected a client because the IP address and port number do not match.
16#8003	Not all data could be sent. Remaining data will be sent in the next cycle(s).
16#8004	Not all data received: Received length < Expected length

ERROR = TRUE

Error code	Meaning	Error only for
16#C001	Socket creation failed.	
16#C002	IP has wrong format.	
16#C003	Memory allocation failed.	
16#C100	Unexpected error during connecting of a client to a server.	TCP/TLS_SOCKET
16#C101	Unexpected error during receive operation.	UDP/TCP/TLS_RECEIVE
16#C102	Unexpected error during send operation.	UDP/TCP/TLS_SEND
16#C103	Unexpected error during bind operation.	UDP_SOCKET
16#C104	Unexpected error during listen operation.	TCP/TLS_SOCKET
16#C105	Unexpected error during accept operation.	TCP/TLS_SOCKET
16#C150	The TLS parameterization of the TLS_SEND/TLS_RECEIVE function blocks is inconsistent with the TLS_SOCKET function block. This is the case when: • TLS_SEND/TLS_RECEIVE require secure transmission/reception of data (SEND_SECURE/RECEIVE_SECURE = TRUE), but the socket is not yet initialized for TLS communication (START_TLS = FALSE). • TLS_SEND/TLS_RECEIVE require insecure transmission/reception of data (SEND_SECURE/RECEIVE_SECURE = FALSE), but the socket is already initialized for TLS communication (START_TLS = TRUE).	TLS_~
16#C151	An error regarding the START_TLS input of the TLS_SOCKET function block has occurred. START_TLS was set from TRUE to FALSE during opened TLS socket (ACTIVE = TRUE).	TLS_SOCKET

Error code	Meaning	Error only for
16#C201	There are too many open sockets in the underlying socket provider.	
16#C202	An operation on a nonblocking socket cannot be completed immediately.	
16#C204	The datagram is too long.	
16#C205	Only one use of an address is normally permitted. In case of a TCP or TLS connection, this error code can be emitted when a rising edge is detected at the <code>ACTIVATE</code> input while the <code>ACTIVE</code> and <code>BUSY</code> outputs both are still not <code>FALSE</code> (i.e., a new connection is requested while the previous socket termination is not yet completed). This error also occurs if the controller is switched to <code>STOP</code> and afterwards to the <code>RUN</code> state as this terminates established connections. This error is also emitted if a TCP/TLS server shall listen to several clients. To avoid this error use the newer FB generation <code>TLS_~_2</code>.	
16#C206	The selected IP address is not valid in this context.	
16#C207	The connection was aborted by the .NET Framework or the underlying socket provider.	
16#C208	The connection was reset by the remote peer.	
16#C210	The application tried to send or receive data, and the socket is not connected.	
16#C211	No such host is known. The name is not an official host name or alias.	
16#C212	An internal error occurred with unclear reason. On controllers up to firmware 2019.3 one cause for this error is an empty CipherList. Another known cause is trying to send a UDP datagram to a broadcast address (255.255.255.255) using the <code>UDP_SEND</code> or <code>UDP_SEND_2</code> function block. In case the error is thrown for the <code>UDP_SEND_2</code> FB, ensure that the gateway in your IP configuration is set to a valid address (address 0.0.0.0 is invalid and treated as a missing gateway). (It is not required that the network contains a device with the configured gateway address.)	
16#C213	The remote host is actively refusing a connection. The service is not available on the remote host.	
16#C214	- Parameters like CipherList, TrustStoreName and IdentityStoreName are invalid or missing. - An invalid port number was specified.	
16#C215	A blocking operation is in progress.	
16#C216	The overlapped operation was aborted due to the closure of the <code>System.Net.Sockets.Socket</code> .	

Error code	Meaning	Error only for
16#C217	The application has initiated an overlapped operation that cannot be completed immediately.	
16#C218	A blocking System.Net.Sockets.Socket call was canceled.	
16#C219	An attempt was made to access a System.Net.Sockets.Socket in a way that is forbidden by its access permissions.	
16#C21A	An invalid pointer address was detected by the underlying socket provider.	
16#C21B	A System.Net.Sockets.Socket operation was attempted on a non-socket.	
16#C21C	A required address was omitted from an operation on a System.Net.Sockets.Socket.	
16#C21D	An unknown, invalid, or unsupported option or level was used with a System.Net.Sockets.Socket.	
16#C21E	The protocol type is incorrect for this System.Net.Sockets.Socket.	
16#C21F	The protocol is not implemented or has not been configured.	
16#C220	The support for the specified socket type does not exist in this address family.	
16#C221	The address family is not supported by the protocol family.	
16#C222	The protocol family is not implemented or has not been configured.	
16#C223	The address family specified is not supported. This error is returned if the IPv6 address family was specified and the IPv6 stack is not installed on the local machine. This error is returned if the IPv4 address family was specified and the IPv4 stack is not installed on the local machine.	
16#C224	Network is down.	
16#C225	Network unreachable.	
16#C226	Network dropped connection on reset.	
16#C227	No free buffer space is available for a System.Net.Sockets.Socket operation.	
16#C228	A request to send or receive data was disallowed because the Socket has already been closed.	
16#C229	The connection attempt timed out, or the connected host has failed to respond.	
16#C22A	The operation failed because the remote host is down.	
16#C22B	There is no network route to the specified host. Could not connect to DEST_IP.	
16#C22C	Too many processes are using the underlying socket provider.	
16#C22D	The network subsystem is unavailable.	
16#C22E	The version of the underlying socket provider is out of range.	

Error code	Meaning	Error only for
16#C22F	The underlying socket provider has not been initialized.	
16#C230	A graceful shutdown is in progress.	
16#C231	The specified class was not found.	
16#C232	The name of the host could not be resolved. Try again later.	
16#C233	The error is unrecoverable or the requested database cannot be located.	
16#C234	The requested name or IP address was not found on the name server.	

11.6 FILE_OPEN

Error ID	Meaning
0	No error information available.
2	The maximum number of files is already open.
4	The file is already open.
5	File is write protected or access denied.
6	File name not defined.

11.7 FILE_SEEK

Error ID	Meaning
0	No error information available.
1	Invalid file handle.
13	Invalid positioning mode or position specified is before the beginning of the file.
24	Position could not be set.

11.8 FILE_READ

Error ID	Meaning
0	No error information available.
1	Invalid file handle.
10	End of data reached.
12	The number of characters to be read is greater than the data buffer.
22	No data could be read.

11.9 FILE_CLOSE

Error ID	Meaning
0	No error information available.
1	Invalid file handle.
20	File could not be closed.

11.10 STRING_TO_BUF

Status	Meaning
0	The copy process has been finished correctly.
1	The VAR_IN_OUT descriptors used for the parameter SRC and BUFFER are invalid. This is an internal error.
2	The length of the source buffer does not fit. The size of bytes to be copied assigned in BUF_CNT is larger than the available size of the SRC.
3	The length of the destination buffer does not fit. The sum of the bytes to be copied assigned in BUF_CNT and the offset in the connected byte stream assigned in BUF_OFFS is larger than the size of the connected byte stream.
4	This data type is not supported.
5	The alignment does not fit to this data type. The size to be copied assigned in BUF_CNT must be divisible without remainder by the size of the data type.
6	The conversion INTEL/MOTOROLA (Little-/Big-Endian) has failed.
7	The string length does not fit. Additional checks are necessary for the data type string.
8	The destination buffer has a wrong data type. In some cases the data type is checked. This is described in the special chapter for each data type.
9	The offset value is not correct. In some cases the offset is checked. This is described in the special chapter for each data type.
10	The BUF_CNT does not fit. In some cases the size to be copied is checked.
11	The addresses of the source and the destination are the same.

12 Third-party licenses

This software makes use of third-party packages with different licenses.

The software packages and the licenses are listed in this chapter.

Software (package) name	License type
Software A	MIT license
Software B	BSD license
Software C	GPL license

The corresponding license texts are available in the text file **license.txt** on the library page within the PLCnext Store, under the “Documents” section.

13 Known issues

Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. At vero eos et accusam et justo duo dolores et ea rebum. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet. Lorem ipsum dolor sit amet, consetetur sadipscing elitr, sed diam nonumy eirmod tempor invidunt ut labore et dolore magna aliquyam erat, sed diam voluptua. At vero eos et accusam et justo duo dolores et ea rebum. Stet clita kasd gubergren, no sea takimata sanctus est Lorem ipsum dolor sit amet.

14 Startup example(s)

For the startup instructions of the library function blocks, please find the following examples:

- LIBID_EXA_Name.pcwex
- ...

14.1 Description of LIBID_EXA_Name.pcwex

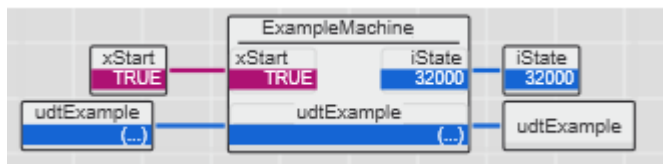
This startup example demonstrates ...

Required hardware:

Item designation	Item no.	Firmware version
AXC F 2152	2404267	2026.0 LTS
AXL SE RS232	1181787	

PLCnext Engineer version: 2026.0 LTS

For starting the ExampleMachine function block, *xStart* input must be set to TRUE.



For the source code, please open the associated PLCnext Engineer project.

14.2 Description of ...

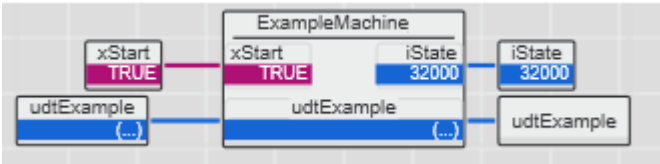
This startup example demonstrates ...

Required hardware:

Item designation	Item no.	Firmware version
AXC F 2152	2404267	2026.0 LTS
AXL SE RS232	1181787	

PLCnext Engineer version: 2026.0 LTS

For starting the ExampleMachine function block, *xStart* input must be set to TRUE.



For the source code, please open the associated PLCnext Engineer project.

15 Support

For technical support, please contact: Company_Name

E-Mail: support@your-company.com

In case of a support request, we need:

- Development system with:
 - Name (e.g., PC Worx, PLCnext Engineer)
 - Version (e.g., PLCnext Engineer 2022.0.1 LTS)
- Bus structure / plant including all articles with
 - Name
 - Order number
 - Firmware version
 - External components
- Used libraries with:
 - Name
 - Version (e.g., IOL_Basic_7)